Telemetry Protocol Specification

Protocol: LoRa telemetry
Protocol version: 1
Frame size: 14 bytes
Transport: SX1276 / RadioLib
Byte order: Little-endian for multibyte integer fields

1. Purpose and scope

The protocol transfers telemetry from an ESP32-based transmitter to a receiver over a LoRa radio link. A frame contains a protocol identifier, protocol version, message type, telemetry sequence number, POST status, temperature, humidity, configuration version, and frame CRC.

This document specifies the implemented frame format. It does not imply that acknowledgement, retransmission, encryption, or receiver-to-transmitter feedback is implemented.

2. Frame structure

Every telemetry frame is exactly 14 bytes long.

Offset	Size	Field	Type	Description
0	1 byte	Magic	`uint8`	Fixed value `0xA5`; identifies the frame
1	1 byte	Protocol version	`uint8`	Current value: 1
2	1 byte	Message type	`uint8`	Value 1 = telemetry
3	2 bytes	Telemetry sequence	`uint16`	Sequence counter, little-endian
5	2 bytes	POST failure mask	`uint16`	Bitmask of failed power-on self-tests, little-endian
7	2 bytes	Temperature	`int16`	Temperature in tenths of a degree Celsius, little-endian

9	2 bytes	Humidity	`uint16`	Relative humidity in tenths of a percent, little-endian
11	1 byte	Configuration version	`uint8`	Low 8 bits of the active configuration version
12	2 bytes	Frame CRC	`uint16`	CRC16-CCITT over bytes 0–11, little-endian

Total: 14 bytes.

2.1 Magic, version, and message type

- `Magic = 0xA5` is the frame marker.
- `Protocol version = 1` identifies this layout.
- `Message type = 1` identifies a telemetry frame.

A receiver should reject a frame if the magic, protocol version, message type, or CRC is invalid.

2.2 Telemetry sequence

The sequence field is a 16-bit unsigned integer. Its bytes are transmitted least-significant byte first. The receiver can use the value to identify repeated frames and observe gaps in received telemetry.

The counter wraps around after `65535`. Sequence comparison must account for wraparound.

2.3 POST failure mask

The POST failure mask reports failures detected by the transmitter's power-on self-test.

Bit	Mask	Meaning
0	`0x0001`	NVS initialization or readiness failure
1	`0x0002`	DHT sensor test failure
2	`0x0004`	Radio initialization failure

3	0x0008	Configuration validation failure
4–15	—	Reserved; currently unused

A bit value of 1 means the corresponding check failed. A value of 0 means that the check did not report a failure.

POST failure mask = 0x0000 means that none of the four implemented POST checks reported a failure. It does not guarantee that every component will continue to operate correctly after startup.

The radio POST check verifies initialization, not successful end-to-end packet delivery.

2.4 Temperature

Temperature is encoded as a signed 16-bit integer in units of 0.1 °C.

Example: 237 represents 23.7 °C.

The byte order is little-endian. A receiver should decode the 16-bit value as a signed integer before dividing by 10.

The sentinel 0x7FFF indicates an invalid or unavailable temperature measurement. When this sentinel is present, the receiver must not interpret the value as a real temperature.

2.5 Relative humidity

Humidity is encoded as an unsigned 16-bit integer in units of 0.1 % RH.

Example: 654 represents 65.4 % RH.

The byte order is little-endian. The sentinel 0xFFFF indicates an invalid or unavailable humidity measurement.

2.6 Configuration version

Byte 11 carries the least-significant 8 bits of the active configuration version. This field is distinct from the configuration sequence counter stored in NVS.

The transmitted field is only one byte, even if the configuration version in firmware is a wider integer.

3. CRC

The frame CRC is CRC16-CCITT, calculated over bytes 0 through 11 inclusive.

- Initial value: `0xFFFF`
- Polynomial: `0x1021`
- Input data: first 12 bytes of the frame
- CRC field: bytes 12–13
- CRC byte order: little-endian

The receiver recalculates the CRC over the first 12 bytes and compares it with the transmitted value. A mismatch means the frame must be rejected.

The CRC detects many accidental transmission or data-corruption errors. It does not provide cryptographic authentication or protection against deliberate modification.

4. Message types and payloads

Type	Value	Payload
Telemetry	`0x01`	Sequence, POST mask, temperature, humidity, configuration version

Only telemetry message type `0x01` is defined in this version.

No acknowledgement, configuration-write, command, or diagnostic response frame is defined by this specification.

5. Versioning and compatibility

The protocol version is carried in byte 1.

A receiver supporting version 1 should parse frames using the 14-byte layout in this document. Unknown protocol versions should be rejected rather than parsed using the version 1 layout.

Changes to field offsets, field sizes, byte order, CRC parameters, or the meaning of existing fields require a documented compatibility decision and, where necessary, a new protocol version.

The configuration version in byte 11 is not the protocol version. The two fields serve different purposes.

6. Errors, acknowledgement, retries, and timeouts

6.1 Receiver-side validation

The receiver should validate, in order:

1. Frame length is exactly 14 bytes.

2. Magic is `0xA5`.
3. Protocol version is supported.
4. Message type is supported.
5. CRC matches the received frame.

Frames failing validation should not be accepted as valid telemetry.

6.2 Acknowledgements and retries

The current telemetry protocol does not define an application-level ACK frame. The transmitter does not receive confirmation that the remote receiver decoded a packet.

A successful local radio transmit call indicates that the local transmit operation completed; it does not prove that the remote receiver received or accepted the frame.

Application-level retry policy and ACK timeout are not specified or implemented in protocol version 1.

6.3 Error reporting

The transmitter reports local initialization, POST, and transmission status through its serial log. The POST failure mask is also included in telemetry frames.

A receiver should log invalid frames and radio receive errors for test diagnosis. Receiver-side error reporting does not constitute feedback to the transmitter.

7. Radio limits and operating parameters

The frame payload is 14 bytes. Radio airtime also depends on LoRa modulation settings, preamble, coding rate, bandwidth, spreading factor, and radio configuration.

The implementation initializes the SX1276 using the configured frequency and RadioLib defaults unless other parameters are explicitly set in firmware. The exact modulation settings must be confirmed from the firmware and test logs before publishing measured airtime or throughput.

This specification does not claim a measured maximum packet rate, link range, packet delivery ratio, or maximum supported payload.

The operating frequency and transmit power must comply with the applicable radio regulations in the deployment location. The configured frequency must be verified before field testing.

8. Example frame and decoding

The following is an illustrative example of a valid frame layout, not a captured test result:

```
A5 01 01 2A 00 00 00 ED 00 8E 02 02 [CRC_L] [CRC_H]
```

Decoded fields before CRC verification:

- Magic: `0xA5`
- Protocol version: 1
- Message type: telemetry (1)
- Sequence: 42
- POST failure mask: `0x0000`
- Temperature: `0x00ED` = 237 = `23.7 °C`
- Humidity: `0x028E` = 654 = `65.4 % RH`
- Configuration version: 2

`[CRC_L] [CRC_H]` are placeholders, not literal bytes. A real captured frame must include the actual CRC bytes and must pass CRC verification before being treated as valid.

For the final test report, replace this illustrative example with a real 14-byte hexadecimal frame captured from the receiver's serial log. Include the raw bytes and the receiver's decoded output.

9. Implementation notes

- The frame uses fixed offsets and a fixed length; it does not contain a separate length field.
- Multibyte frame fields are little-endian.
- The CRC covers the first 12 bytes, including the magic, version, type, and payload fields.
- Invalid sensor readings use sentinel values rather than ordinary measurements.
- Configuration persistence and its NVS CRC are separate from the telemetry frame CRC.